

CLASS 04 / SEP 15

MODULE 1 — FIELDS ARRAYS

PARAMETRIC BEGINS HERE

0.1 + 0.2 DUE

1.1 ISSUED TODAY / ARRAYS + ATTRACTORS

WHERE WE JUST MOVED TO

Module 0 is finished. The ground has changed.

PROCEDURAL

MODULE 0 — DONE

A fixed sequence of operations.
You defined the steps.
One expected result.

PARAMETRIC

MODULES 1-3 — NOW

Relationships between inputs and form.
You define the rules of variation.
A family — you know the range,
not each member.

An array on its own is still procedural — repeat this N times. It becomes parametric the moment a parameter governs how the copies differ.

AN ARRAY IS A LIST OF POSITIONS

Not a command that duplicates things. A list you compute, then place geometry at.

INDEX → POSITION

$i = 0 \rightarrow p = \text{start}$

$i = 1 \rightarrow p = \text{start} + \text{step}$

$i = 2 \rightarrow p = \text{start} + 2 \times \text{step}$

$i = 3 \rightarrow p = \text{start} + 3 \times \text{step}$

$p[i] = \text{start} + i \times \text{step}$

YOU ALREADY KNOW THIS

Move is addition. An array is move, repeated, with the amount driven by the index.

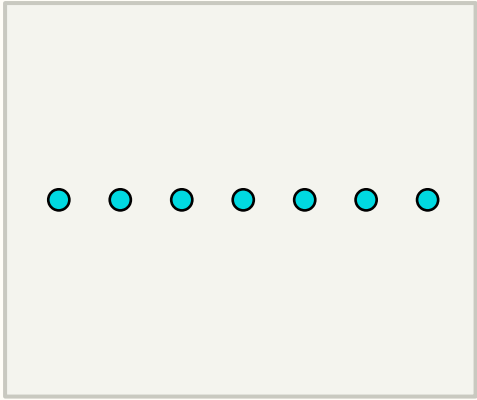
The index is the only new idea: a counter that tells each copy how far along the sequence it is — and therefore how different it should be.

The index is what makes variation possible. Without it every copy is identical, because nothing distinguishes one from another.

FOUR ARRAYS

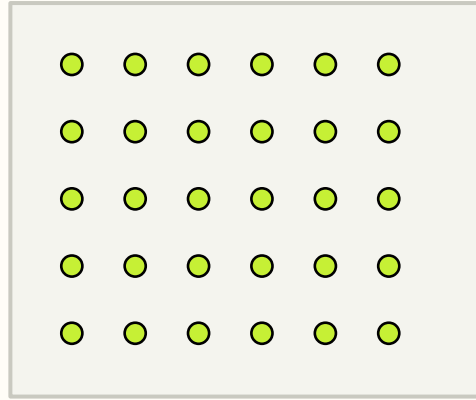
Same idea, four ways of generating the list.

LINEAR



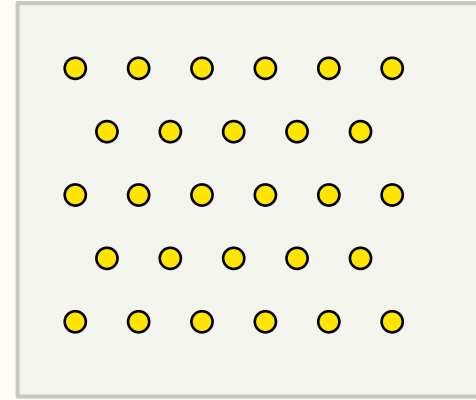
$$p[i] = p_0 + i \cdot d$$

GRID



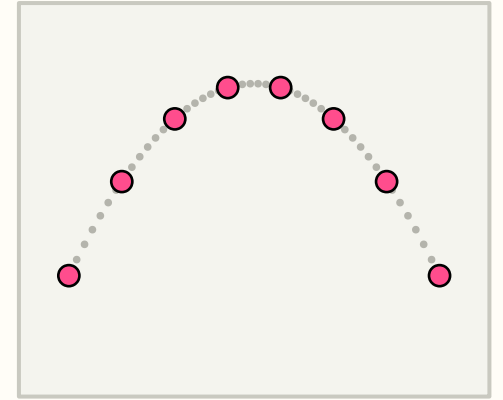
$$p[i][j] = p_0 + (i \cdot dx, j \cdot dy)$$

HEX



offset odd rows by $dx/2$

ON CURVE



$$p[i] = \text{curve at } t = i/n$$

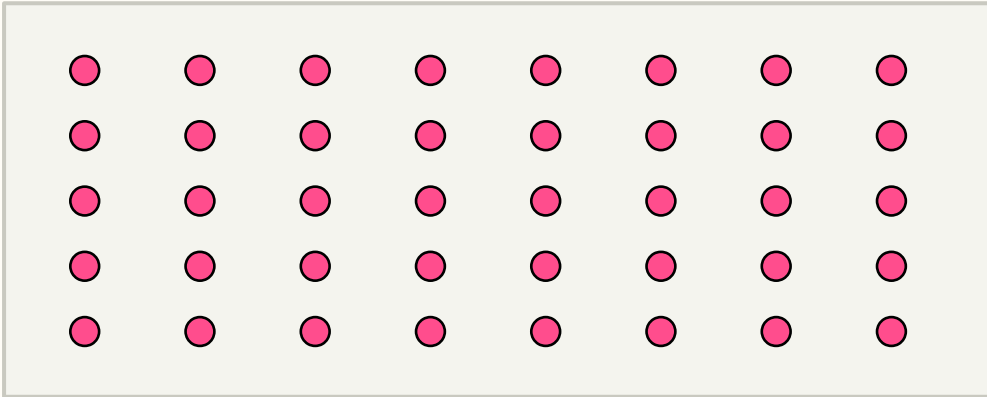
A grid is a list of lists — two counters instead of one. Hold that thought; it returns in Module 3 as a panel field.

REPETITION IS NOT THE POINT

The same grid, twice. One is a pattern. The other is a system.

REPETITION

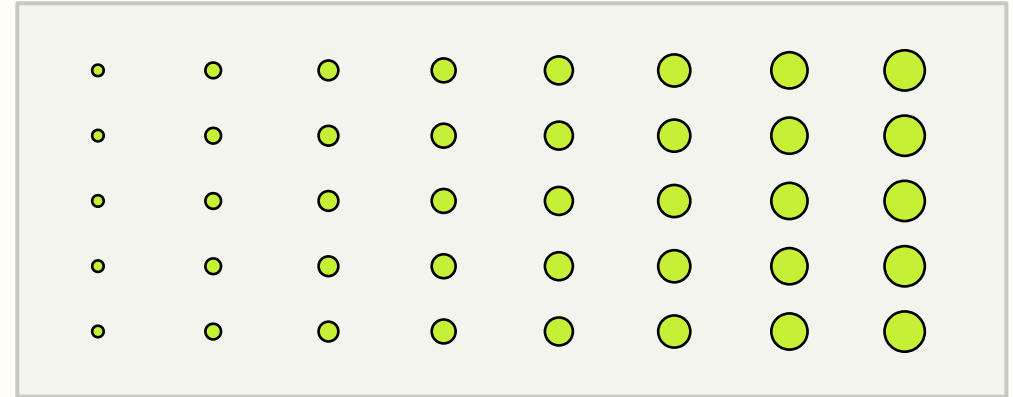
PROCEDURAL



Every element identical. The index is generated and then ignored.

VARIATION

PARAMETRIC

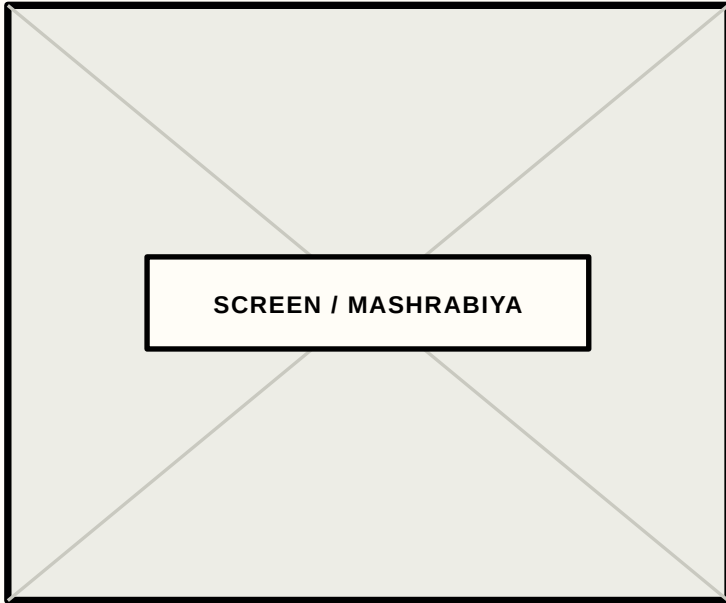


Size driven by the index. The same list, now doing work.

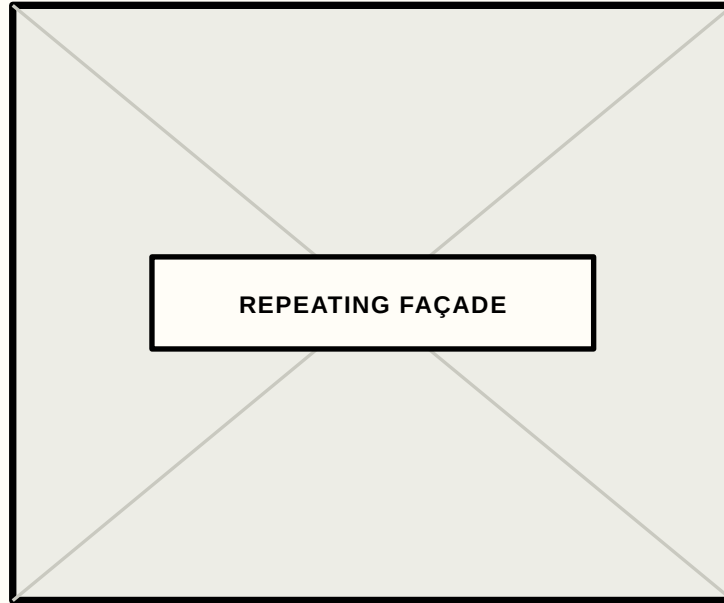
1.1 REQUIRES VARIATION. A FIELD OF IDENTICAL ELEMENTS DOES NOT MEET THE ASSIGNMENT.

ARRAYS IN ARCHITECTURE

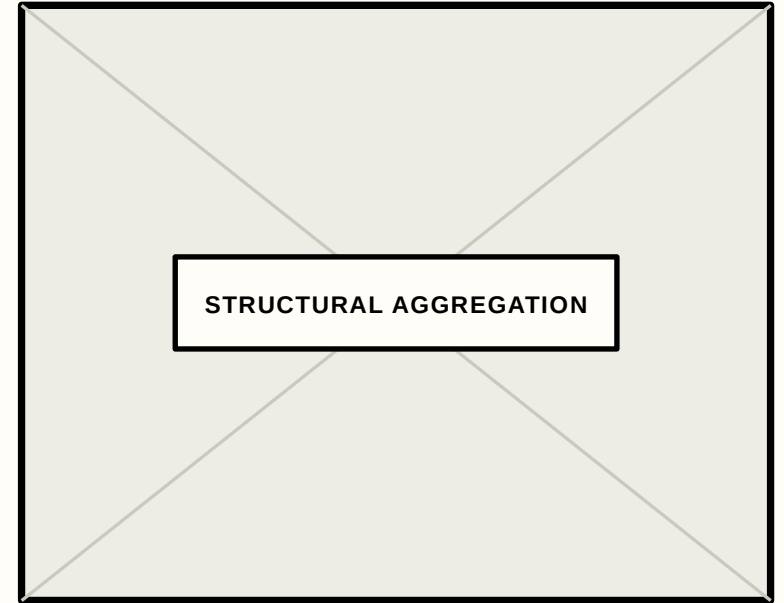
Screens, façades, structure, aggregation. Look for the rule, not the image.



Add: a perforated screen with graded aperture.



Add: a façade whose module varies across the elevation.



Add: a structure built from one repeated element.

FOR EACH ONE, ASK THREE QUESTIONS

What is the module? What generates the positions? What varies across the field, and what drives it?

ISSUED TODAY

1.1 — ARRAYS + ATTRACTORS

DUE CLASS 6 — SEP 29 / 8%

An array organizes points. An attractor differentiates them. This assignment is about what happens when you combine the two — a matrix of approaches, where different array types are driven by different attractor types.

	SINGLE POINT	MULTIPLE POINTS	CURVE
LINEAR			
GRID			
HEX			
ON CURVE			

PICK 3 CELLS.
ANY THREE.

REQUIREMENTS

Counting these earns a C. The rest is judgement.

3+

array/attractor combinations, drawing on all array types taught

3

attractor types across the set: single point, multiple points, curve

3+

parameters driven by attractor distance — scale, rotation, density, height, colour

1,000+

elements in at least one composition

1+

composition resolved spatially in 3D

3 VIEWS

of one combination: isometric, close-up, eye-level

1

precedent analysis — 2–3 sentences: module, positions, what varies

Arrays and falloff authored. Array add-ons, scatter tools and preset falloff plugins are prohibited.

NOW — BUILD THE ARRAYS

Rest of the session: all four array types, working, from one starting point.

✓

Linear array with a controllable count and spacing

✓

Grid array — two counters, not one array duplicated

✓

Hexagonal array — odd rows offset by half the spacing

✓

Array along a curve you drew yourself

✓

One property driven by the index, so the set is not uniform

THE LAST ONE IS THE ASSIGNMENT. THE OTHERS ARE PRACTICE.

BEFORE NEXT CLASS

WATCH

Tutorial 1.2 — attractors. Single point, multiple points, curve. Distance, falloff, remapping. Sent tonight.

BRING

Your working arrays. Next week we drive them with attractors, so bring a file you can build on.

BRING

Your 0.1 procedure, printed. Next class you build a classmate's object from theirs.

LOOK

One precedent for 1.1. Be ready to name the module, the positions, and what varies.